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THE EFFECT OF GDP BENCHMARK REVISIONS ON SOVEREIGN BOND PRICES IN EMERGING MARKET ECONOMIES

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The Effect of GDP Benchmark Revisions on Sovereign Bond Prices in Emerging Market Economies[1]

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Abstract

This paper estimates how sovereign bond yields respond when low- and middle-income countries release a GDP benchmark revision. Using a panel of 16 episodes between 2011 and 2022, it finds that yields fall relative to the cross-country baseline two to three trading days after a release, with the largest decrease at three days when yields sit roughly 2 percent below baseline. The release day itself shows no detectable change, and yields return to the cross-country path within two trading weeks. The response does not scale with the size of the implied change in the debt-to-GDP ratio. Markets appear to price the release as a generally positive signal rather than to weigh it against the arithmetic of debt sustainability.

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1 Introduction

In September 2024, an audit ordered by Senegal's newly elected government revealed the previous administration had concealed the true state of the country's public finances to the degree of nearly \$7 billion in undisclosed borrowing (Court of Auditors of Senegal, 2025). As a result, Senegal revised its debt-to-GDP ratio at the end of 2023 to approximately 100%, compared to 74% that had been earlier reported (International Monetary Fund, 2024*b*). The International Monetary Fund (IMF) soon suspended a \$1.8 billion loan program (International Monetary Fund, 2024*a*). Yields on Senegal's 2033 dollar bonds surged toward record highs (Bloomberg News, 2025). A fiscal crisis had begun.

Instead of following the IMF's calls to restructure its sovereign debt, Senegal's finance ministry announced something unusual. It would conduct a benchmark revision of its GDP: a comprehensive re-estimation of the economy's nominal size using updated data sources and a new 2021 base year (CNBC Africa, 2025). The rationale was clear. If the benchmarking led to a larger nominal GDP number, the Senegalese government could mechanically reduce the debt-to-GDP ratio that had become the focal point of investor anxiety. In response to this news, Senegal's bond yields fell by as much as 31 basis points (Bloomberg News, 2025). Barclays estimated that the revision could lift Senegal's 2024 nominal GDP by 15 to 25 percent, potentially reducing the debt burden they estimated at above 120% to closer to below

100% of GDP (CNBC Africa, 2025). Markets were reacting to an accounting exercise that changes no real economic activity, retires no debt, and creates no new fiscal capacity.

Sovereign debt sustainability is almost universally assessed relative to GDP (International Monetary Fund, 2021). A benchmark revision that pushes GDP upward mechanically improves the debt-to-GDP ratio without any change in real fiscal conditions. In the Senegal episode, the media treated it as a market-moving adjustment, with larger revisions implying larger improvements to a country's fiscal profile. Other benchmarking events fit this narrative. Following Kenya's 2014 benchmark revision, the World Bank highlighted that the country's debt-to-GDP ratio had fallen from 52% to roughly 43%, and estimated that the improved fiscal profile could save Kenya on the order of $10 million annually on a transaction like its $2 billion Eurobond debut (World Bank, 2014). Nigeria's 2025 revision was framed in similar terms by analysts and development institutions as improving the country's creditworthiness in the eyes of global lenders (Punch Newspapers, 2025). Whether markets in fact consistently price these benchmark revisions, and if so, whether the response scales with the size of the revision, is an open question.

The Senegal episode that motivates this paper is, however, an unusual case. Senegal's bond market reaction occurred before any revised GDP data was published, in the context of an acute sovereign debt crisis, an IMF program suspension, and yields already at record highs. In that environment, the prospect of a mechanically lower debt-to-GDP ratio carried immediate, crisis-relevant information for investors already focused on Senegal's fiscal sustainability. Most benchmark revisions, however, are not crisis events. They are technical statistical releases by national statistics offices. They are the product of multi-year data compilation processes, generally in countries not experiencing acute fiscal stress at the time of release. Whether the market reacts more generally to these more routine releases is the question this paper investigates.

Using a panel of 16 GDP benchmark revision episodes across low- and middle-income countries between 2011 and 2022, this study estimates the effect of benchmark revision data releases on 10-year local currency sovereign bond yields. Findings show that yields decrease relative to the cross-country baseline two to three trading days after the release, with the response peaking on the third day, where yields are approximately 2 percent below the pre-release baseline. The release day itself shows no detectable yield change, and the decrease appears to dissipate by the end of the first trading week. Long-horizon estimates show no persistent effect beyond two trading weeks. The market response to a benchmark release is therefore localized in time, concentrated in a narrow post-release window rather than producing a permanent shift in the country's relative yield level. The data do not show a statistically significant scaling of this response with the size of the implied change in the debt-to-GDP ratio. The interaction between the release indicator and the implied ratio change is indistinguishable from zero at every horizon, in both the daily and average specifications. With 16 events, however, the confidence intervals on the interaction are wide, and the result is best understood as the absence of a detectable graded response in this sample rather than as evidence that no such response exists.

These findings paint a picture of how markets process benchmark revisions outside crisis contexts. Yields fall following the release, though the decrease does not persist beyond the first trading week. Pre-release yield changes show no evidence that markets anticipated the release. Moreover, results are consistent with markets treating the act of conducting a benchmark revision as a generally positive event rather than as an input to the arithmetic of debt sustainability. Because all 16 revisions in the sample are positive, the findings should be understood as applying to upward revisions specifically.

This paper makes two contributions. Empirically, it provides cross-country evidence on sovereign bond market reactions to GDP benchmark revision data releases in low- and middle-income countries. Conceptually, it documents a gap between common media reactions to benchmark revisions and the way markets actually price these events. These findings suggest markets react positively to benchmark revisions, independent of the implied change in the debt-to-GDP ratio.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on sovereign bond yields, macroeconomic data releases, and data quality in developing economies. Section 3 provides institutional background on GDP benchmark revisions and the channels through which they may affect sovereign bond markets. Section 4 describes the data. Section 5 presents the empirical methodology. Section 6 reports the results. Section 7 concludes.

2 Literature Review

Two sets of literature are important to this paper. One studies the fiscal determinants of sovereign bond yields and concludes that the debt-to-GDP ratio is among the strongest predictors of borrowing costs in emerging markets. The other studies the quality of GDP measurement in low- and middle-income countries and documents that benchmark revisions to the underlying GDP series can be very large. Neither literature has asked whether the events that explicitly revise the GDP denominator of the debt-to-GDP ratio, while leaving the real economy unchanged, are priced by sovereign bond markets. This paper is the first to do so.

2.1 Sovereign Bond Yields and Fiscal Fundamentals

A long literature establishes that fiscal fundamentals drive sovereign bond yields in developing economies, but this literature treats statistics as accurate and asks how variation in real fiscal conditions affects yields, leaving open whether mechanical changes to the GDP denominator, with no change in real fiscal conditions, are priced.

Edwards (1984) shows that yields on developing-economy sovereign debt are related to the debt-to-GDP ratio, the debt service ratio, and the country's reserve position. Bondholders assess the probability that a sovereign will repay, and fiscal variables are the primary observable signal of repayment capacity. Eaton and Gersovitz (1981) provide the theoretical mechanism. They model a sovereign that borrows up to the point where the benefits of additional consumption smoothing are offset by the expected cost of default, and show that the equilibrium borrowing limit is a function of the sovereign's income, which in practice is estimated by GDP. The model creates a direct theoretical link between measured output and sovereign creditworthiness. Anything that changes the measured level of GDP changes the inferred borrowing capacity of the sovereign, and therefore the yield investors demand.

Subsequent empirical work has refined these results. Baldacci, Gupta and Mati (2008) examine a panel of emerging market economies and find that fiscal consolidation significantly reduces sovereign yields, attributing the result to a fundamentals effect and a fiscal governance signal. Min (1998) finds that the debt-to-GDP ratio, the current account balance, and inflation explain a substantial share of the cross-sectional variation in developing country yields.

This literature establishes the foundation for the question this paper takes up. If the debt-to-GDP ratio is a primary determinant of sovereign borrowing costs, then anything that mechanically changes the GDP denominator could be priced by bond markets, even in the absence of any change in real fiscal conditions. Whether markets in fact treat such mechanical changes the same way they treat changes driven by real fiscal conditions is a question this literature has not asked, because it has not examined the events that produce mechanical changes in isolation. The benchmark revision is exactly such an event.

2.2 GDP Data Quality and Sovereign Risk

A second literature documents that GDP measurement in low- and middle-income countries is often unreliable and that benchmark revisions can be very large, but it has not connected these observations to financial-market responses, leaving open whether the release of a benchmark revision moves sovereign bond yields.

Jerven's *Poor Numbers* (2013) is the central reference. Drawing on a detailed examination of national accounts compilation practices across sub-Saharan Africa, Jerven documents that many countries have gone decades between benchmark revisions, relying on base years from the 1980s or earlier to extrapolate current economic activity. The result is that official GDP figures may substantially understate the true size of the economy, particularly in sectors that have grown rapidly since the last benchmark. When a benchmark revision finally occurs, the revision can be enormous. Ghana's 2010 benchmark revision raised measured GDP by 60%, and Nigeria's 2014 revision raised it by 82% (Jerven, 2013; Devarajan, 2013). A country that benchmarks, after going decades without doing so, is announcing that all prior fiscal sustainability assessments were made against potentially severely inaccurate data.

A related strand of work examines the relationship between institutional quality and sovereign borrowing costs. Baldacci, Gupta and Mati (2008) find that political risk indicators affect yields in emerging markets. The implication is that markets price the quality of the institutions that produce fiscal data, not just the fiscal data themselves.

These two strands together imply that the release of a benchmark revision should carry market-relevant information of two kinds. First, the revision changes the GDP denominator of the debt sustainability ratio. Second, the act of conducting a revision is itself a signal about the country's investment in its statistical infrastructure. Yet no prior work has examined how sovereign bond markets respond to benchmark revision releases specifically. Existing event-study work on macroeconomic announcements (Andritzky, Bannister and Tamirisa, 2007) has examined rating changes and IMF program announcements but not benchmark revisions. The gap this paper fills is the absence of any evidence on whether the events that explicitly revise the GDP denominator of the debt sustainability ratio move bond markets, and whether the size of the revision shapes that response.

3 Institutional Background

3.1 GDP Benchmarking and Rebasing

National accounts estimates of gross domestic product can be understood as the sum of prices multiplied by quantities across all measured sectors of the economy. In a country that has N sectors, nominal GDP in any given year is computed as $\sum_{s=1}^{N} p_s^t \cdot q_s^t$, where p_s^t is the current price of sector s in year t, q_s^t is the measured quantity of output in that sector, and N is the number of sectors included in the national accounts framework. Real GDP holds prices fixed at a designated base year, using p_s^{base} in place of p_s^t, so that changes over time reflect movements in quantities alone. In practice, statistical offices construct real GDP using a variety of index methodologies, including Laspeyres, Paasche, Fisher, and chain-weighted approaches, each of which handles the choice of reference prices differently. For the purposes of this paper, the relevant point is that rebasing affects only the construction of real GDP and leaves nominal GDP unchanged, regardless of which index methodology is used.

In a benchmarking year, the national statistics office comprehensively remeasures both prices and quantities across all sectors through censuses, enterprise surveys, household surveys, and administrative data compilation, often in coordination with the central bank and with technical assistance from the IMF or World Bank (Silungwe, 2020). This re-measurement updates the values for both p_s^t and q_s^t. In the years between benchmarks, the statistics office does not repeat this comprehensive measurement. Instead, it extrapolates the benchmark-year quantities forward using partial indicators (Silungwe, 2020). While this approach ensures continuous GDP estimates, the quality of the estimates progressively deteriorates. The indicators may not cover all sectors, and the further the economy moves from the conditions captured in the base year, the less reliable the extrapolated quantities become.

Two distinct processes are involved in a comprehensive revision that addresses this deterioration. The first is rebasing, which updates the fixed prices p_s^{base} by switching the base year to a more recent period. This affects real GDP but does not change nominal GDP, since nominal GDP uses current prices rather than base year prices. The second process, benchmarking, updates the measured quantities q_s and can also expand the number of sectors N by incorporating new comprehensive survey and census data (Silungwe, 2020). Benchmarking, unlike rebasing, can change nominal GDP values. In practice, national accounts programs typically undertake both activities as part of one process, and often misname the joint revision as rebasing, when in fact they are also benchmarking (Silungwe, 2020). Table 1 summarizes which inputs each process updates and how each affects real and nominal GDP. The resource intensity of the benchmarking process means that undertaking one is itself a substantive commitment to data quality.

Take Nigeria's 2014 revision as an example. The rebasing component updated p_s^{base} by switching the base year from 1990 to 2010. The benchmarking component expanded N to include rapidly growing new sectors like telecommunications, the Nollywood film industry, and e-commerce, and collected new data on the quantities q_s produced in each. After adding so many new sectors during the benchmarking exercise, Nigeria's revised GDP was approximately 80% higher than previously calculated.

Table 1: What Rebasing and Benchmarking Change

	Rebasing	Benchmarking
Base-year prices (p_s^{base})	Updates	—
Current-year quantities (q_s^t)	—	Updates
Number of measured sectors (N)	—	May expand
Real GDP	Affected	Affected
Nominal GDP	Not affected	Affected

Notes: Rebasing updates the reference prices used to construct constant-price (real) GDP without changing the measured quantities of output, leaving nominal GDP unchanged. Benchmarking re-measures current quantities and may expand the number of sectors covered, changing both real and nominal GDP. National accounts revisions in practice typically combine both processes (Silungwe, 2020). The distinction matters for sovereign debt analysis because the debt-to-GDP ratio uses nominal GDP, which only benchmarking can change.

The inputs that a benchmarking exercise can update are visible directly in the GDP identity. Treating measured nominal GDP $Y_t = \sum_{s=1}^{N} p_s^t \cdot q_s^t$ as a function of its inputs, the change in Y produced by simultaneous updates to prices, quantities, and the count of measured sectors is

$$dY = \sum_{s=1}^{N} (q_s \, dp_s + p_s \, dq_s) + p_{N+1} q_{N+1}, \tag{1}$$

where the first sum collects the contributions from price and quantity updates within the existing frame and the final term captures the contribution of a newly admitted sector when the frame expands from N to $N+1$. Take a hypothetical economy in which the statistical office measures three sectors A, B, and C, and a fourth sector D exists in the economy but sits outside the measurement frame. A benchmark revision can re-measure the prices and quantities of A, B, and C, contributing the price and quantity terms in Equation (1), and can admit D to the frame, contributing the additional term $p_D q_D$. A benchmark revision can therefore change the published GDP figure through any combination of price updates, quantity updates, and additions to the measurement frame.

The change in measured GDP passes through directly to the debt-to-GDP ratio. Let B_t denote the stock of outstanding sovereign debt and consider the ratio B_t/Y_t. Holding the debt stock fixed, the partial derivative of the ratio with respect to measured GDP is

$$\frac{\partial}{\partial Y_t}\left(\frac{B_t}{Y_t}\right) = -\frac{B_t}{Y_t^2}. \tag{2}$$

A benchmark revision changes Y_t but not B_t, so the implied change in the ratio is

$$d\left(\frac{B_t}{Y_t}\right) = \frac{\partial}{\partial Y_t}\left(\frac{B_t}{Y_t}\right)\cdot dY_t = -\frac{B_t}{Y_t^2}\,dY_t. \tag{3}$$

The ratio falls when $dY_t > 0$, but the rate at which it falls shrinks as Y_t grows. The same dollar of upward revision lowers the ratio by less when the measured GDP is already high. This curvature, together with the linear scaling in B_t, means the same proportional revision can imply quite different ratio changes across countries. Whether markets price the ratio change, and whether the response tracks dY_t or the implied ratio change itself, is what Section 5 tests.

Benchmark revisions typically produce upward adjustments to measured GDP (World Bank, 2024). Downward revisions are possible given the right combination of methodological adjustments. The structural reason for the upward tendency is developed in Section 3.2. The empirical pattern matters for interpreting this paper's results: all 16 revisions in the sample are positive, and the findings should be read as applying to upward revisions specifically.

The benchmarking process itself typically takes one to three years from initiation to public release. The public release is generally made through a non-routine press release or statistical bulletin issued by the national statistics office. The question of whether markets could partially anticipate a benchmark revision before the release is relevant to the identification strategy and is addressed in Section 5.

3.2 What a Benchmark Revision Measures

Benchmark revisions in low- and middle-income countries skew positively. Section 3.1 described what a benchmarking exercise updates in the national accounts framework. To make precise what the resulting revision number represents, the study decomposes published GDP into its underlying real value and the measurement error introduced by the statistical process.

Let Y_t^* denote the true nominal value of economic activity in year t. This quantity is unobservable. The statistical office produces a measured estimate Y_t that differs from the truth by a measurement error ε_t:

$$Y_t = Y_t^* + \varepsilon_t. \tag{4}$$

True nominal GDP aggregates prices times quantities across every sector that exists in the economy. Let N^* denote the true number of sectors. Then

$$Y_t^* = \sum_{s=1}^{N^*} p_s^t \cdot q_s^t. \tag{5}$$

The statistical office measures $N \leq N^*$ sectors, so measured GDP is

$$Y_t = \sum_{s=1}^{N} p_s^t \cdot q_s^t. \tag{6}$$

Subtracting Equation (6) from Equation (5) gives the coverage component of the measurement error:

$$\varepsilon_t^c \equiv - \sum_{s=N+1}^{N^*} p_s^t \cdot q_s^t. \tag{7}$$

The coverage error is weakly negative, $\varepsilon_t^c \leq 0$, by construction. Prices and quantities are nonnegative for every sector, so a sector outside the measurement frame can only be missed entirely. Missing economic activity always understates GDP. The statistical office cannot mismeasure a sector that is not in its frame in a way that overstates output.

The remaining component of measurement error captures every other source of difference between true and measured GDP within the frame. The residual is referred to as the methodology error and is denoted as ε_t^m. By construction, it absorbs any mismeasurement of the prices and quantities of measured sectors, along with any analytical choices the statistical office makes in compiling the accounts. Examples of the latter include the valuation of owner-occupied housing, the choice of deflation index, sectoral classification rules, and the imputation procedures used for the informal sector. Methodology error is sign-unrestricted, since the underlying inputs can deviate from the truth in either direction. The total measurement error is therefore

$$\varepsilon_t = \varepsilon_t^c + \varepsilon_t^m, \tag{8}$$

and measured GDP can be written as $Y_t = Y_t^* + \varepsilon_t^c + \varepsilon_t^m$.

A benchmark revision at time τ expands the measurement frame from N to N' sectors, where $N < N' \leq N^*$, and updates price, quantity, and methodology inputs. The revised estimate Y_τ' is a new application of the same identity, with an updated coverage error $\varepsilon_\tau^{c\prime}$ and an updated methodology error $\varepsilon_\tau^{m\prime}$:

$$Y_\tau' = Y_\tau^* + \varepsilon_\tau^{c\prime} + \varepsilon_\tau^{m\prime}. \tag{9}$$

The new coverage error is

$$\varepsilon_\tau^{c\prime} = -\sum_{s=N'+1}^{N^*} p_s^\tau \cdot q_s^\tau. \tag{10}$$

The expanded frame, $N' > N$, means the sum in Equation (10) is taken over fewer omitted sectors than the sum in Equation (7). The new coverage error is therefore weakly closer to zero than the old one, $\varepsilon_\tau^{c\prime} \geq \varepsilon_\tau^c$. Benchmarking weakly reduces the magnitude of coverage error.

Let ρ_τ denote the observed revision size, defined as the difference between the revised and original estimates,

$$\rho_\tau \equiv Y_\tau' - Y_\tau, \tag{11}$$

and substituting the two GDP expressions causes the Y_τ^* terms to cancel:

$$\rho_\tau = \underbrace{(\varepsilon_\tau^{c\prime} - \varepsilon_\tau^c)}_{\text{coverage component} \geq 0} + \underbrace{(\varepsilon_\tau^{m\prime} - \varepsilon_\tau^m)}_{\text{methodology component, sign unrestricted}}. \tag{12}$$

The first term is weakly positive by the argument above. The second term has no sign restriction, since a methodology update can move measured GDP in either direction depending on the specific changes adopted. Two consequences follow.

The real economy at the moment of release does not enter the revision size. Y_τ^* cancels in the derivation of Equation (12), so whatever the magnitude of

ρ_τ, it carries no information about how large the underlying economy actually is at the time of the revision. The revision size reflects only the corrections to past measurements.

The asymmetry between the two components of the revision provides a structural account of the empirical regularity noted in Section 3.1. The coverage component is weakly positive by construction. The methodology component is sign-unrestricted and can push in either direction. Their sum, therefore, has a built-in asymmetric pressure toward upward revisions: the coverage term never pulls down and can pull strongly up. In contrast, the methodology term can move either way. The pull from the coverage term is largest in the settings where benchmarks are most overdue, since the omitted-sector sum in ε_t^c has had a long interval to accumulate. New sectors that did not exist in the prior frame enter N^* but not N, and their contribution to GDP accumulates outside the measurement frame until the benchmark unwinds it. Downward revisions can arise when the coverage component is small, because the prior frame was already broadly current, and the methodology component is negative, because new methodology valuations or classifications produced lower aggregates. The cross-event pattern in this paper's sample, where all 16 events in the regression panel are positive, is consistent with the structural asymmetry derived above (Section 4).

3.3 Benchmark Revisions versus Routine Revisions

It is worth noting that not all changes to GDP estimates are benchmark revisions. National accounts data are subject to continuous revisions as more data become available. Preliminary estimates based on partial data are successively revised as final administrative records, tax data, and survey results are incorporated (World Bank, 2024). Benchmark revisions are categorically different. They go beyond updating the numbers within an existing national accounts framework; they change the framework itself.

3.4 Channels Through Which Markets May Respond

A benchmark revision can affect bond markets through several channels. Three are plausible given the prior literature and the institutional context: a data-credibility channel, a political-capacity channel, and a fiscal-ratio channel. The first two operate through the categorical fact of the revision independent of the revision size. The third operates through the mechanical change in the debt-to-GDP ratio and predicts a graded response. This subsection lays out all three.

The data-credibility channel is forward-looking. Conducting a benchmark revision expands the statistical office's measurement frame, which mechanically reduces the magnitude of coverage error in the most recent estimate and shrinks the set of unmeasured sectors from which future coverage error can accumulate. The signal investors receive is that subsequent estimates will rest on a more complete frame than prior estimates did. The Brookings Institution describes this channel in detail, arguing that benchmarking signals a country is emphasizing the reliability of its statistics, which can improve investor confidence (Sy, 2015).

The political-capacity channel reads the revision as a signal about the state of the country. Conducting a benchmark revision is resource-intensive and requires sustained coordination across the statistical office, the central bank, and external technical assistance providers. A country that completes such an exercise reveals institutional capacity and political will to invest in its statistical infrastructure. The literature reviewed in Section 2 documents that political risk and institutional quality are priced in sovereign bonds (Baldacci, Gupta and Mati, 2008), so a signal of this kind is a plausible direct input to bond prices.

The fiscal-ratio channel is different in that the size of the revision matters. When a benchmark revision raises measured nominal GDP, the denominator of the debt-to-GDP ratio increases while the numerator, the stock of outstanding debt, remains unchanged. The ratio mechanically decreases, and the country's fiscal position appears stronger. Under this mechanism, the size of the yield response should scale with the size of the revision, since a larger revision produces a larger improvement in the ratio. Whether it holds is what the empirical specification in Section 5 is designed to test.

These three channels are not mutually exclusive, and the market response to a given benchmark revision reflects the combined effect of whichever are operative. The empirical question this paper asks is narrower than disentangling them. It asks first whether markets price benchmark revision data releases at all, and second whether the response scales with the size of the revision in the way the fiscal-ratio channel would imply.

4 Data

The IMF publishes the World Economic Outlook (WEO) twice annually, and each vintage contains the latest GDP estimate for every member country. Comparing a country's GDP for a given data year across consecutive vintages reveals when a substantial revision occurred. To identify which of these revisions correspond to formal benchmarking exercises, National Statistics Office (NSO) publications are used to determine the exact date when the results of the benchmark revision were released. These dates were then confirmed through contemporaneous news coverage and IMF records. Revisions that do not reflect formal benchmarking are excluded.

Sixteen episodes across 15 countries form the estimation sample. These events do not exhaust the universe of GDP benchmark revisions over the period. They merely represent the events for which corresponding bond data at a daily frequency were available. The Philippines is the only country that appears twice, with benchmarking episodes in 2011 and 2020. Revision sizes range from 4.8 percent (Philippines, 2020) to 82.2 percent (Nigeria, 2014), with a median of 15.5 percent and a mean of 19.4 percent. Table 2 summarizes the events. Figure 1 shows the distribution of revision sizes across the 16 events in the estimation sample, and Figure 2 shows the distribution of the implied changes in the debt-to-GDP ratio. Figure 3 shows the geographic distribution of the 16 countries in the estimation sample.

Table 2: GDP Benchmark Revision Events in the Estimation Sample

Country	Date	Δ GDP	Δ Debt/GDP
Philippines	May 2011	+5.8%	−5.4%
Zambia	Mar 2014	+19.7%	−16.5%
Nigeria	Apr 2014	+82.2%	−45.1%
Kenya	Sep 2014	+21.9%	−18.0%
Tanzania	Dec 2014	+31.4%	−23.9%
Indonesia	Feb 2015	+4.9%	−4.6%
Thailand	May 2015	+8.3%	−7.7%
Turkey	Dec 2016	+19.7%	−16.4%
Ghana	Sep 2018	+25.4%	−20.3%
Georgia	Nov 2019	+8.6%	−7.9%
Vietnam	Dec 2019	+26.4%	−20.9%
Philippines	Apr 2020	+4.8%	−4.6%
South Africa	Aug 2021	+11.0%	−9.9%
Bangladesh	Nov 2021	+16.6%	−14.2%
Pakistan	Jan 2022	+14.4%	−12.6%
Morocco	Apr 2022	+8.7%	−8.0%

Notes: Δ GDP is the percentage change in nominal GDP between consecutive WEO vintages for the same data year. Δ Debt/GDP is the percentage change in the debt-to-GDP ratio implied by the revision, holding the stock of outstanding debt fixed. Events are sorted chronologically by release date.

The dependent variable is the daily change in 10-year sovereign bond yields, measured using Refinitiv Eikon local-currency-denominated government bond indices. The estimation sample combines daily yield observations for all 16 sample countries over 2010–2025 with the release indicator activated only on the relevant release date for each event. This panel construction al-

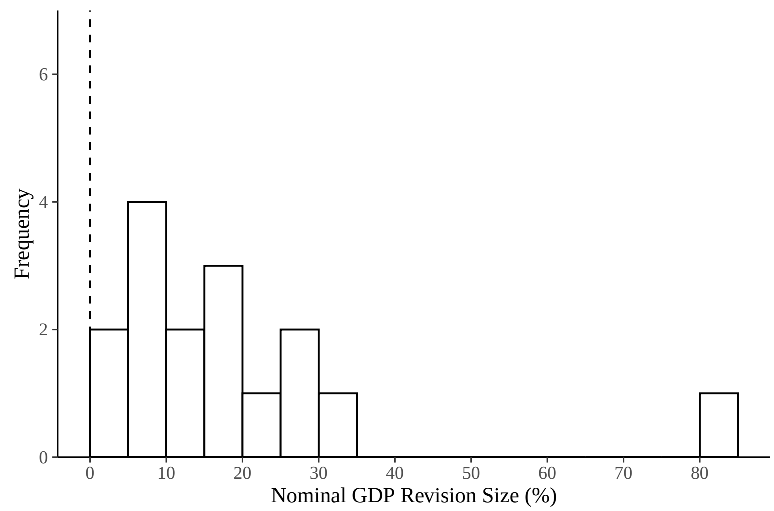

Figure 1: Distribution of GDP Benchmark Revision Sizes

Notes: Distribution of nominal GDP revision sizes across the 16 benchmark episodes in the estimation sample. The dashed line marks zero. Nigeria (2014) is the rightmost observation at +82.2%. All revisions in the estimation sample are positive.

lows the date fixed effects in the regression specification to absorb common global shocks identified off the full cross-section of countries on each calendar date, rather than off the sparse overlap that would result from restricting the sample to event windows alone. The treatment effect is therefore identified from the deviation of the treated country's yield change at horizon h relative to the cross-country average yield change on the same date, after absorbing country-level fixed effects.

The use of local currency bond indices is a limitation of the data. Local currency sovereign bond markets in low- and middle-income countries are generally less liquid and less widely traded than hard currency (typically U.S. dollar-denominated) sovereign bond markets, and the dependent variable in-

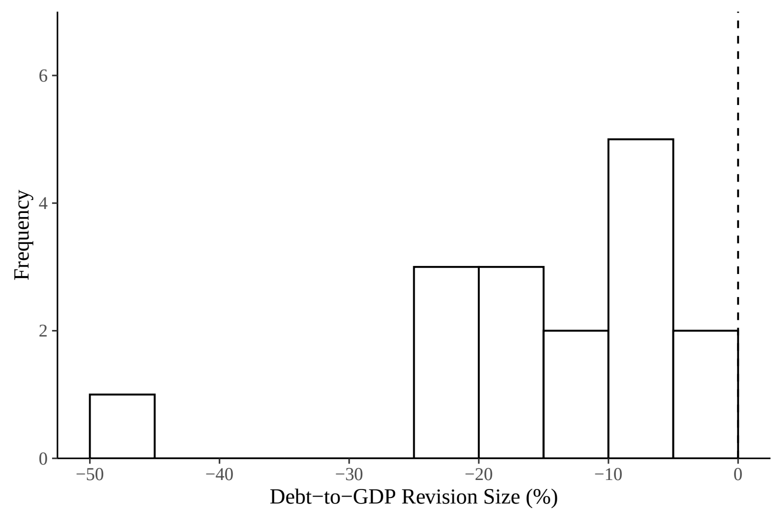

Figure 2: Distribution of Implied Debt-to-GDP Changes

Notes: Distribution of implied changes in the debt-to-GDP ratio across the 16 benchmark episodes in the estimation sample, computed as the percentage change in the ratio holding the stock of outstanding debt fixed. The dashed line marks zero. Nigeria (2014) is the leftmost observation at −45.1%. All implied changes in the sample are negative, as all GDP revisions in the sample are positive.

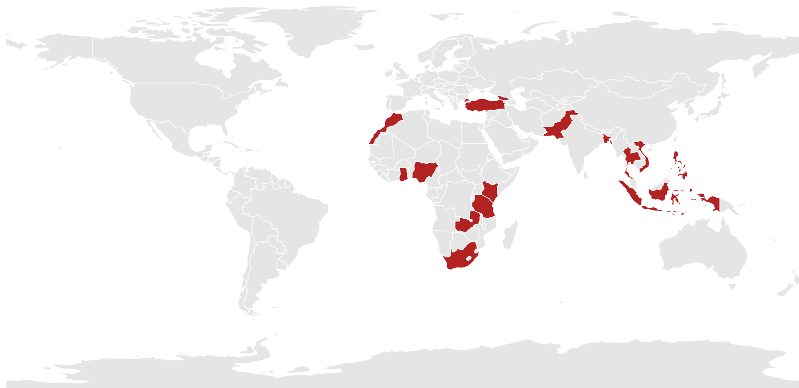

Figure 3: Geographic Distribution of Sample Countries

Notes: Highlighted countries are the 15 low- and middle-income countries in the estimation sample, spanning five regions: Sub-Saharan Africa, the Middle East and North Africa, South and Southeast Asia, and the Caucasus. The Philippines contributes two benchmark events; all other countries contribute one each.

herits whatever measurement noise this thinner trading produces. The choice reflects data availability rather than design; hard currency 10-year bond data are not consistently available across the full sample at the daily frequency required for the event study.

5 Empirical Methodology

This section describes the empirical framework used to estimate the effect of GDP benchmark revision releases on sovereign bond yields. First, this section presents the main estimating equation and defines each of its components. Then, it discusses the fixed effects structure, the economic logic of the interaction term, the approach to statistical inference, and the robustness checks performed.

5.1 Model Specification

The empirical strategy estimates whether sovereign bond yields respond to GDP benchmark revision data releases and whether the magnitude of the response varies with the size of the revision in the debt-to-GDP ratio. For each horizon h,

$$\Delta y_{i,t}^{(h)} = \beta_0 D_{i,t}^{(h)} + \beta_1 \left(D_{i,t}^{(h)} \times \delta_i \right) + \alpha_i + \alpha_t + u_{i,t} \tag{13}$$

where i indexes benchmark revision events, t indexes calendar dates, and h denotes the horizon over which the yield change is measured. An event consists of a treated country and a ± 30 calendar day window centered on the release date of its benchmark revision. Within this window, the treated country contributes daily yield observations alongside daily observations from all other sample countries; the indicator $D_{i,t}^{(h)}$ equals one on the trading day h days after the release date for the treated country in event i and zero on all other country-date observations. The subscript i corresponds to individual benchmark episodes rather than countries, since the Philippines appears twice in the sample (2011 and 2020). The dependent variable $\Delta y_{i,t}^{(h)}$ is an h-day yield change, defined precisely in Section 5.2. δ_i is the change in the country's debt-to-GDP ratio implied by the benchmark revision, expressed as a decimal. Holding the stock of outstanding debt fixed, an upward GDP revision mechanically lowers the debt-to-GDP ratio, so δ_i is negative for all events in the sample. α_i and α_t are event and date fixed effects, respectively. Standard errors are clustered at the event level.

The two coefficients of interest are β_0 and β_1. The coefficient β_0 captures the average yield change at horizon h across all events, independent of the change in δ_i. The coefficient β_1 captures how the yield response scales with the implied change in debt-to-GDP. The fiscal-ratio story that underlies the common media reaction implies $\beta_1 > 0$, since a more negative δ_i (a larger improvement in the ratio) should produce a more negative yield change.

5.2 Dependent Variable

The dependent variable $\Delta y_{i,t}^{(h)}$ is an h-day log change in the 10-year local currency sovereign bond yield index, anchored at the yield on the trading day before the release. Results for $h \in \{1, 2, 3, 4, 5\}$ trading days are reported using two specifications. In both, the indicator $D_{i,t}^{(h)}$ equals one on the trading day h days after the release date for the treated country in event i and zero on all other country-date observations. Persistence at longer horizons of $h \in \{10, 15, 20, 30\}$ trading days is reported separately in Section 6.4.

The first is a daily response. For horizon h, the dependent variable is $\log(y_{i,t_0+h}) - \log(y_{i,t_0-1})$, where t_0 is the release date. This is the cumulative log yield change from one trading day before the release through day $t_0 + h$, measuring whether yields by day h have moved relative to the pre-release baseline.

The second is an average response. For horizon h, the dependent variable is $\frac{1}{h}\sum_{j=1}^{h}\left[\log(y_{i,t_0+j}) - \log(y_{i,t_0-1})\right]$. This is the average across horizons 1 through h of the daily response, putting all post-release trading days on equal footing rather than weighting day $t_0 + h$ alone.

Anchoring both specifications at $t_0 - 1$ rather than the release date itself preserves the full release-day yield change as part of the response, which is necessary because a release that occurs after the trading day's close cannot be priced until the following trading day. The daily specification at horizon h measures whether the yield at the end of trading day $t_0 + h$ differs from the pre-release baseline; the average specification asks whether yields across the entire post-release window differ from that baseline. Reporting both allows the data to reveal whether a market response is concentrated near the release or spread across the post-release window.

5.3 Fixed Effects Structure and Identification

The identification strategy uses two sets of fixed effects that together absorb a large portion of the variation in sovereign bond yields unrelated to the benchmark revision.

Event fixed effects α_i absorb all time-invariant characteristics of each benchmark episode. Such characteristics include the country's baseline yield level, the structure and liquidity of its local currency bond market, and the political context surrounding the revision. The implied change in debt-to-GDP δ_i is itself constant within an event and is therefore absorbed by α_i, which is why it enters Equation (13) only through the interaction with $D_{i,t}^{(h)}$.

Date fixed effects α_t absorb global market conditions on each calendar date. Any factor that is common across all sample countries on a given day is captured by α_t, including risk sentiment shifts, US monetary policy announcements, and commodity price movements. This is important because the event windows span a decade and overlap with large and small global market shocks that affect emerging market yields.

After absorbing both sets of fixed effects, the remaining identifying variation comes from within-event, within-date deviations in yield changes between the treated country at horizon h and other sample countries on the same day. The identification assumption is parallel trends: conditional on event and date fixed effects, the average yield change for the treated country at each horizon h following the release would have moved in parallel with the cross-country average in the absence of the benchmark revision. This is a counterfactual statement about post-release dynamics and is fundamentally untestable. The pre-trends test reported in Section 6.1 examines whether treated yields moved differentially in the days before the release; finding no pre-trends does not establish parallel trends but lends indirect credibility to the assumption.

5.4 Parallel Trends

The parallel trends assumption stated in Section 5 requires that, in the absence of the benchmark revision, the treated country's yield path would have moved in parallel with the cross-country average after the release date. This counterfactual cannot be tested directly. As an indirect check, the paper examines whether yields moved differentially in the days preceding the release.

Benchmark revisions are typically the product of one to three years of data compilation by the national statistics office, often with technical assistance from the IMF or World Bank. While market participants may know that a revision is in progress, the precise release date is rarely fixed in advance and is generally announced through a press release or statistical bulletin issued on short notice. The exact magnitude of the revision is not knowable until the moment of release. These features make the release a credible information shock, even though the existence of the revision process is itself public knowledge.

If markets had been pricing in expected revisions during the run-up to the release, yields would show a downward drift in the days before each event. The absence of such drift does not prove that post-release parallel trends would hold counterfactually, since pre-release behavior cannot be used to verify a statement about the post-release counterfactual; rather, it removes one specific reason to doubt the assumption. Section 6.1 reports estimates of Equation (13) on the five trading days preceding each release, with $D_{i,t}^{(h)}$ replaced by pre-release indicators D_{i,t_0-h} that activate h trading days before the release date and matching backward-looking yield changes on the left-hand side. Coefficients on the pre-release indicators and their interactions with δ_i are statistically indistinguishable from zero at every lead, consistent with no anticipatory yield movements.

5.5 Inference and Clustering

Standard errors are clustered at the event level throughout. The benchmark release varies at the event level, not at the country-day level, and yield changes within a ± 30 day window around a single release are likely correlated through common local factors. Clustering at the event level allows for arbitrary within-event correlation of errors across days in the window.

A limitation of this approach is that the sample contains 16 events, which is below the conventional threshold of 30 to 50 clusters typically recommended for reliable cluster-robust inference (Cameron, Gelbach and Miller, 2008). With few clusters, cluster-robust standard errors can be downward-biased, which could produce over-rejection of the null hypothesis. The average specification produces a stable, negative release-day coefficient at $h = 2$, $h = 3$, and $h = 4$, with t-statistics that exceed conventional thresholds. The consistency of the estimate across these horizons is unlikely to arise from inference artifacts alone.

6 Results

6.1 Parallel Trends

Section 5 described the parallel trends assumption underlying the identification strategy and argued that the institutional features of benchmark revisions make it credible. This subsection reports the empirical pre-trends test that lends indirect support to the assumption.

Equation (13) is estimated on the five trading days preceding each release, with $D_{i,t}^{(h)}$ replaced by pre-release indicators D_{i,t_0-h} that activate h trading days before the release date. The dependent variable is the corresponding backward-looking single-day log change between days t_0-h-1 and t_0-h. If markets had been pricing in the revision in the days leading up to the release, these pre-release coefficients would be statistically distinguishable from zero.

Table 3: Parallel Trends Test

	(1) $h=-1$	(2) $h=-2$	(3) $h=-3$	(4) $h=-4$	(5) $h=-5$
Dummy (β_0)	0.0039	0.0088	−0.0040	0.0186	0.0029
	(0.0037)	(0.0077)	(0.0042)	(0.0164)	(0.0100)
Dummy × δ (β_1)	−0.0086	0.0034	0.0049	0.0353	−0.0051
	(0.0147)	(0.0234)	(0.0252)	(0.0511)	(0.0278)
Event FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is the single-day backward log change in the 10-year local currency sovereign bond yield, $\log(y_{i,t_0-h}) - \log(y_{i,t_0-h-1})$. The indicator D_{i,t_0-h} activates h trading days before the release date. The interaction term uses the change in the country's debt-to-GDP ratio implied by the revision. Cluster-robust standard errors at the event level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Both the dummy and interaction coefficients are statistically indistinguishable from zero at every lead. The point estimates do not show systematic movement in any direction, and the confidence intervals comfortably include zero across the five horizons. There is no evidence that yields move in anticipation of the release, consistent with the institutional argument in Section 5.4 that the precise timing and magnitude of a release are not knowable in advance even when the revision process itself is known to be in progress. As discussed in Section 5.4, this finding does not prove that parallel trends holds in the post-release window, but it removes one specific reason to doubt the assumption.

6.2 Average Yield Response

Table 4 reports the estimates of Equation (13) using the average specification at horizons $h \in \{1, 2, 3, 4, 5\}$ trading days. The dependent variable is the average across horizons 1 through h of the daily log yield change relative to the pre-release baseline, as defined in Section 5.2.

The release indicator coefficient β_0 is negative and statistically significant at $h = 2$, $h = 3$, and $h = 4$, with point estimates of -0.0113, -0.0199, and -0.0164 respectively. The interpretation is that the average daily log yield change over the post-release window is between 1.1 and 2.0 percent lower than the cross-country average on the same dates. The effect strengthens from $h = 2$ to $h = 3$, attenuates slightly at $h = 4$, and is statistically indistinguishable from zero at both $h = 1$ and $h = 5$. The 1-day result reflects that the immediate post-release trading day on its own does not

Table 4: Average Yield Response to GDP Benchmark Revision Releases

	(1) 1-Day	(2) 2-Day	(3) 3-Day	(4) 4-Day	(5) 5-Day
Dummy (β_0)	0.0066	-0.0113^{***}	-0.0199^{***}	-0.0164^{**}	0.0001
	(0.0074)	(0.0033)	(0.0048)	(0.0058)	(0.0108)
Dummy $\times$ δ (β_1)	0.0506	0.0031	-0.0121	0.0120	0.0680
	(0.0376)	(0.0108)	(0.0336)	(0.0366)	(0.0554)
Event FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is the average daily log change in the 10-year local currency sovereign bond yield over the h trading days following the release, anchored at the yield on the trading day before the release. The interaction term uses the change in the country's debt-to-GDP ratio implied by the revision. Cluster-robust standard errors at the event level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

register a detectable yield change, which is examined further in Section 6.3. The 5-day result indicates that, by the end of the first trading week, the average response over the window is no longer detectable.

The interaction coefficient β_1 is statistically indistinguishable from zero at every horizon. The point estimates change sign across horizons and are small relative to their standard errors. The fiscal-ratio prediction described in Section 3.4, which implies $\beta_1 > 0$ because larger improvements in the debt-to-GDP ratio (more negative δ_i) should produce more negative yield changes, is not supported by the data. The interpretation of this null is discussed in Section 6.5.

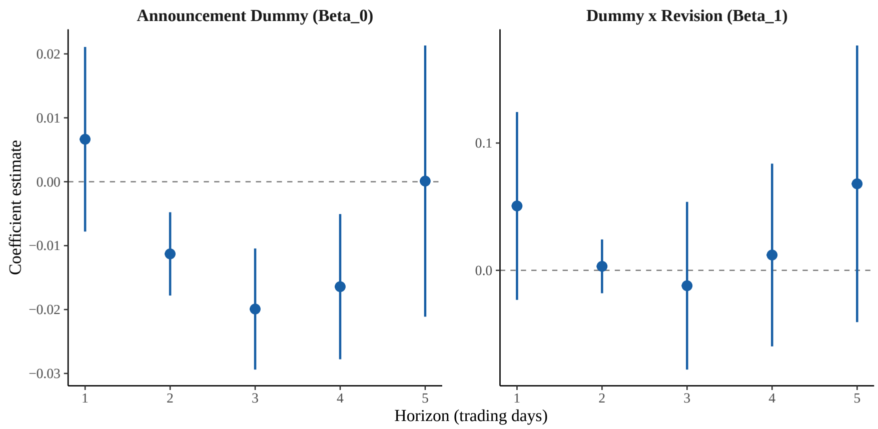

Figure 4: Average specification: release indicator (β_0) and δ interaction (β_1) coefficients across $h = 1$ to $h = 5$ trading days, with 95% confidence intervals.

6.3 Daily Yield Response

Table 5 reports the estimates of Equation (13) using the daily specification at horizons $h \in \{1, 2, 3, 4, 5\}$ trading days. The dependent variable is the cumulative log change in the 10-year local currency sovereign bond yield from the trading day before the release through day $t_0 + h$, as defined in Section 5.2.

The pattern in the daily specification largely mirrors the average specification. The release indicator coefficient β_0 is statistically indistinguishable from zero at $h = 1$, becomes significantly negative at $h = 2$ and $h = 3$ with point estimates of -0.0134 and -0.0260, attenuates to marginal significance at $h = 4$, and is no longer significant at $h = 5$. The largest decrease occurs at $h = 3$, where yields are 2.6 percent lower than the pre-release baseline relative to the cross-country average. The interaction coefficient β_1 is statis-

Table 5: Daily Yield Response to GDP Benchmark Revision Releases

	(1) 1-Day	(2) 2-Day	(3) 3-Day	(4) 4-Day	(5) 5-Day
Dummy (β_0)	0.0066	-0.0134^{***}	-0.0260^{***}	-0.0164^{*}	-0.0103
	(0.0074)	(0.0044)	(0.0069)	(0.0085)	(0.0096)
Dummy $\times$ δ (β_1)	0.0506	0.0104	-0.0149	0.0321	0.0626
	(0.0376)	(0.0193)	(0.0451)	(0.0633)	(0.0602)
Event FE	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is the cumulative log change in the 10-year local currency sovereign bond yield from the trading day before the release to day $t_0 + h$, where t_0 is the release date. The interaction term uses the change in the country's debt-to-GDP ratio implied by the revision. Cluster-robust standard errors at the event level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

tically indistinguishable from zero at every horizon, mirroring the result in the average specification.

The two specifications agree on the basic pattern: yields are not detectably different from the cross-country baseline on the release day itself, then fall over the next two to three trading days, then return to baseline by the end of the first trading week. The daily specification shows this pattern most sharply at $h = 3$, while the average specification, which weights all horizons through h equally, attenuates the magnitude but produces tighter standard errors. The two specifications are complementary rather than competing: the daily result identifies when the response is concentrated, and the average result confirms that the cumulative effect over the window is not driven by a single noisy day.

That the response is detectable two to three trading days after the release rather than on the release day itself is consistent with several features of the institutional setting. Local-currency sovereign bond markets in low- and middle-income countries are less liquid than hard-currency markets, and price discovery in these markets may proceed gradually rather than through a single-day repricing. National statistics offices release benchmark revisions on schedules that do not necessarily align with local trading hours, and a release that occurs after market close cannot be priced until the following trading day. The implications of a benchmark revision for sovereign creditworthiness require analysts to recompute fiscal ratios and update debt sustainability assessments, which is not instantaneous.

6.4 Persistence

The results in Sections 6.2 and 6.3 establish that yields fall relative to the cross-country baseline two to three trading days after the release, and that the average response over the window is no longer detectable by $h = 5$. A natural follow-up question is whether the decrease observed at short horizons leaves any persistent imprint on the yield path, or whether yields return fully to the counterfactual cross-country path within the first trading week. Table 6 reports estimates at horizons $h \in \{10, 15, 20, 30\}$ trading days using the average specification.

The decrease observed in the first trading week does not persist. At $h = 10$, the release indicator coefficient is positive and statistically indistinguishable from zero. At $h = 15$, the coefficient is positive and statistically significant at the 1 percent level, with a point estimate of 0.0048, indicating

Table 6: Persistence of Yield Response Beyond the First Trading Week

	(1) 10-Day	(2) 15-Day	(3) 20-Day	(4) 30-Day
Dummy (β_0)	0.0035	0.0048***	0.0002	-0.0043
	(0.0032)	(0.0012)	(0.0009)	(0.0041)
Dummy $\times$ δ (β_1)	0.0161	0.0135**	0.0026	-0.0655
	(0.0142)	(0.0047)	(0.0035)	(0.0818)
Event FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is the average daily log change in the 10-year local currency sovereign bond yield over the h trading days following the release, anchored at the yield on the trading day before the release. The interaction term uses the change in the country's debt-to-GDP ratio implied by the revision. Sample at long horizons is reduced relative to the main results due to missing yield data: 8 events at $h = 10$, 13 events at $h = 15$, $h = 20$, and $h = 30$. Cluster-robust standard errors at the event level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

that the average daily yield change over the three weeks following a release is slightly above the cross-country baseline rather than below it. By $h = 20$ the coefficient returns to zero, and at $h = 30$ it is negative but not statistically distinguishable from zero. The interaction coefficient β_1 follows the same pattern, significant only at $h = 15$ and indistinguishable from zero at the other three horizons.

These results should be interpreted with caution. The long-horizon sample is reduced by data availability: only 8 events have yield data through $h = 10$, and 13 events have data through $h = 15$, $h = 20$, and $h = 30$. The shrinking sample further weakens the already limited inference described in Section 5. The $h = 15$ result is the only significant long-horizon coefficient, and it is the opposite sign of the short-horizon results, which is unusual.

The most defensible reading of the persistence pattern is that the decrease observed in the first trading week does not extend beyond it. Yields return to the cross-country path within roughly two trading weeks of the release. The market response to a benchmark revision is therefore localized in time rather than producing a permanent shift in the country's relative yield level. This is consistent with a transient information shock. Markets process the release and adjust prices for two to three trading days, by which time other information clouds out the signal from the benchmarking.

6.5 Revision Size

The interaction coefficient β_1 is statistically indistinguishable from zero at every horizon in the main results and in both the daily and average specifications. The point estimates change sign across horizons and are small relative to their standard errors. The data do not show that the yield significantly scales with the size of the implied debt-to-GDP improvement.

This result is informative for distinguishing among the channels described in Section 3. The fiscal-ratio channel that prevails in media reactions predicts $\beta_1 > 0$: a more negative δ_i produces a larger mechanical improvement in the debt-to-GDP ratio, which under that channel should produce a larger yield decrease. The data do not provide evidence for this prediction. The estimated response is consistent with markets pricing the release as a generally positive signal rather than proportional to the implied fiscal improvement.

With 16 events, however, the confidence intervals on β_1 are wide enough at most horizons to admit modest scaling in either direction. The data are most cleanly read as ruling out large-sized effects rather than ruling out any size effect at all. Distinguishing a small but real graded response from a true null would require a substantially larger sample of benchmark events than is currently available at the daily-frequency bond data needed for this design.

The persistence results in Section 6.4 include one significant interaction coefficient at $h = 15$, with a positive sign that would be consistent with the fiscal-ratio channel. Its interpretation is constrained by the same caveats that apply to the level coefficient at the same horizon: it is the only significant interaction in the persistence sample, the sign is the opposite of the short-horizon estimates, and the long-horizon sample is reduced by data availability. This isolated coefficient is not read as decisive evidence in either direction.

Because all 16 revisions in the sample are positive, the result speaks only to upward revisions.

7 Conclusion

This paper estimates the effect of GDP benchmark revision data releases on 10-year local currency sovereign bond yields across 16 episodes in low- and middle-income countries from 2011 to 2022. Yields decrease relative to the cross-country baseline two to three trading days after the release, with the response peaking on the third day where yields are approximately 2 percent below the pre-release baseline. The release day itself shows no detectable yield change, and the decrease dissipates by the end of the first trading week. Long-horizon estimates show no persistent effect beyond two trading weeks. The interaction between the release indicator and the implied change in the debt-to-GDP ratio is statistically indistinguishable from zero at every horizon, in both the daily and average specifications. The market response to a benchmark release is therefore localized in time and does not visibly scale with the size of the implied fiscal improvement.

These findings partially support the prevailing media narrative around benchmark revisions. In the Senegal episode that motivates this paper, bond yields fell sharply on the announcement of a benchmarking exercise, with hopes that the debt-to-GDP ratio would fall. In the 16 examples studied in this paper, markets respond to these releases, but the response is small, short-lived, and unrelated to the magnitude of the implied debt-to-GDP improvement. The fiscal-ratio channel that describes the common narrative around benchmark revisions, in which a larger upward revision implies a larger improvement in fiscal sustainability, is not visible in the data on routine releases. Markets appear to treat the act of conducting a benchmark revision as a generally positive signal rather than as an input to the arithmetic of debt sustainability.

Several features of the data and design constrain the interpretation of these results. First, the sample contains 16 benchmark events. This number falls below conventional thresholds for robust inference, and although the average specification produces a stable, negative release indicator coefficient at horizons of two through four trading days, the confidence intervals on the interaction with revision size are wide. The data are best read as ruling out large-sized effects of revision magnitude rather than ruling out any size effect at all.

Second, all 16 revisions in the sample are positive. As shown in Section 3.2, this makes sense given the structural asymmetry of the benchmarking process. Still, it means the findings apply specifically to upward revisions and cannot speak to how markets might respond to a downward revision. A downward revision may be a shock to markets, and a sample that includes downward revisions would be needed to test whether the findings of this paper extend to revisions of either sign.

Finally, the dependent variable is the 10-year local currency sovereign bond yield. Local currency markets in low- and middle-income countries are less liquid than hard currency markets, and the dependent variable inherits whatever measurement noise this thinner trading produces. The choice reflects data availability rather than design. The investor base for local currency bonds also differs from the investor base for hard currency bonds, and the response of one market need not match the other.

Three extensions follow from the limitations described above. The first is a hard currency replication. Sovereign hard currency bond yields in low- and middle-income countries are more liquid, more widely traded, and follow the investor base most directly relevant to debt sustainability assessments. A version of this design applied to hard currency yields would test whether the response observed in local currency markets generalizes to the markets where global investors actually trade emerging market sovereign risk.

A second extension is a longer sample. As more low- and middle-income countries undertake benchmark revisions in the coming years, the panel of releasable events will grow. A larger sample would tighten the confidence intervals on the interaction with revision size and would allow for credible heterogeneity analysis. It would also include a meaningful number of downward revisions if any occur, which would let the design speak to the symmetry of the market response that this sample cannot address. A larger panel would also strengthen any future structural analysis of the channels through which benchmark revisions affect sovereign borrowing costs.

The findings carry a measured implication for governments considering when and how to undertake a benchmark revision. Borrowing costs do fall after a release, so the exercise produces a small reduction in yields that policymakers can reasonably point to as a benefit of investing in statistical infrastructure. The reduction, however, is short-lived. Yields return to the cross-country baseline within roughly two trading weeks, and no persistent shift in the country's relative borrowing costs follows from the release.

Benchmarking is therefore not a substitute for the fiscal adjustments and debt management actions that drive sovereign creditworthiness over longer horizons. The Senegal case is the clearest illustration. Senegal announced its benchmarking exercise in the middle of an acute debt crisis, with the explicit goal of mechanically lowering its debt-to-GDP ratio. Bond yields fell on the announcement, but the country remains in serious fiscal distress today. Its IMF program is still suspended, and the structural problems that produced the crisis are unaffected by the accounting exercise. A benchmark revision can improve the optics of a country's fiscal position on paper, but it does not retire debt, increase fiscal capacity, or address the underlying conditions that produce a crisis. Policymakers and analysts should treat benchmark releases as one signal among many in assessing sovereign risk, rather than as a remedy for distress.

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